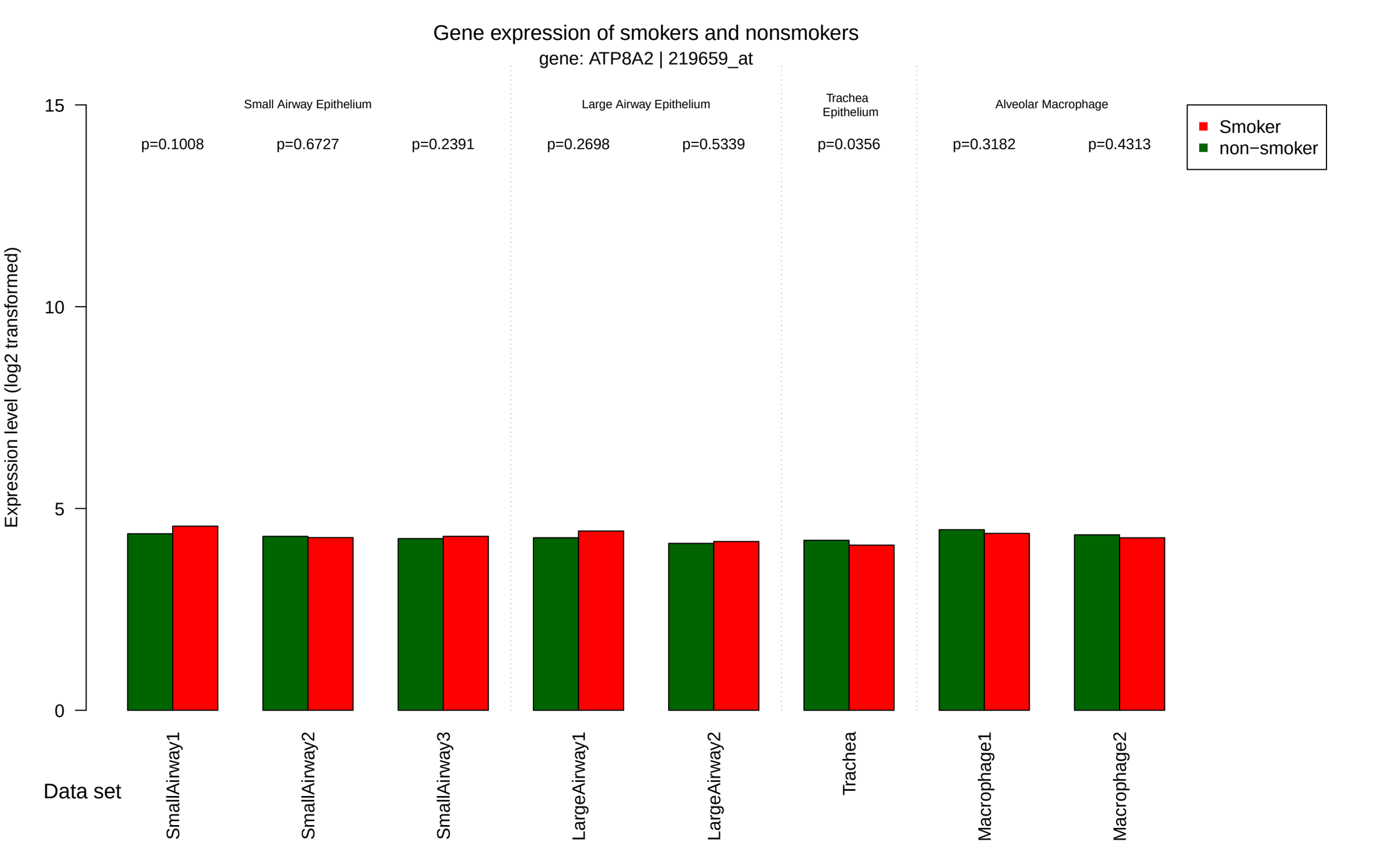
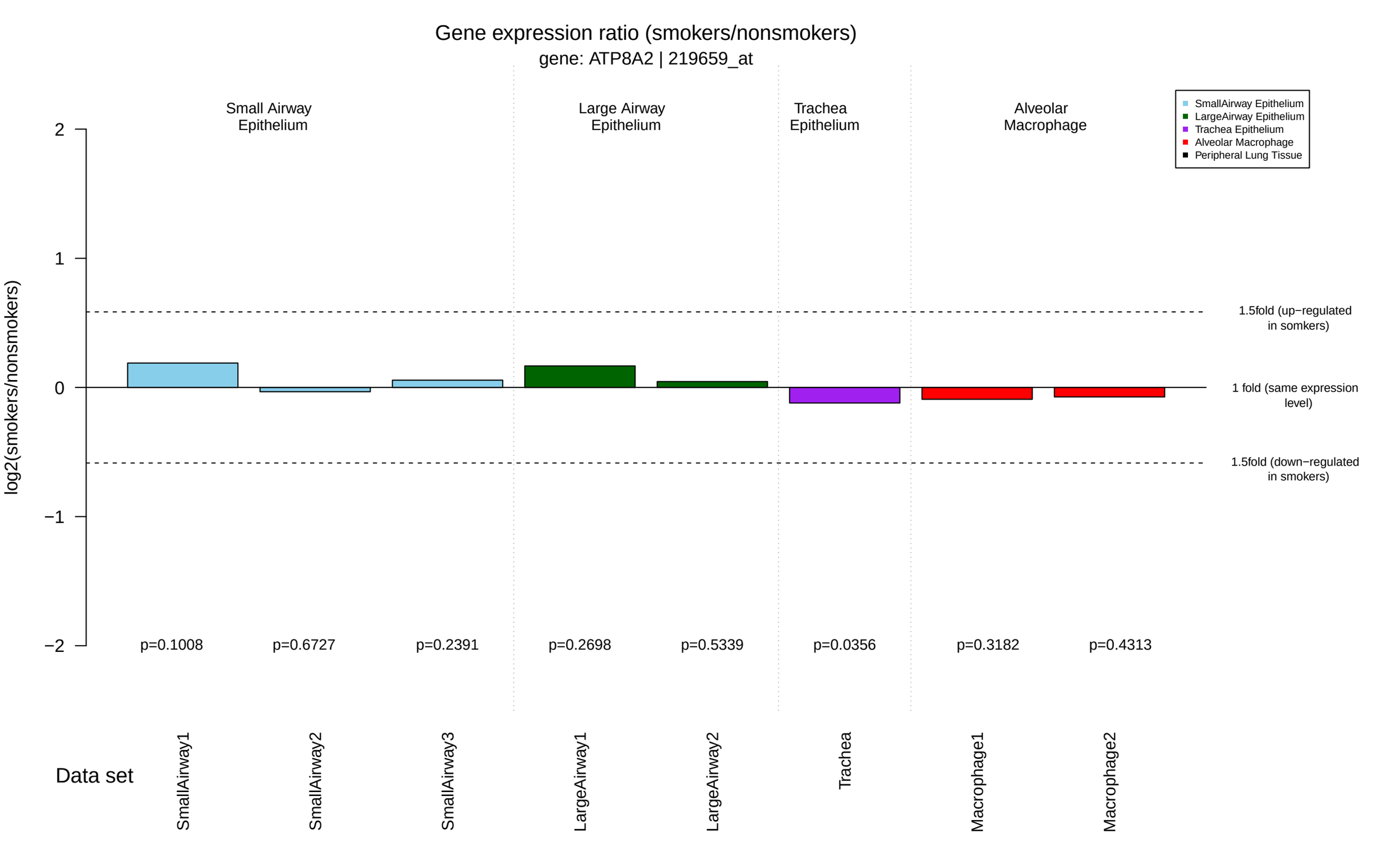
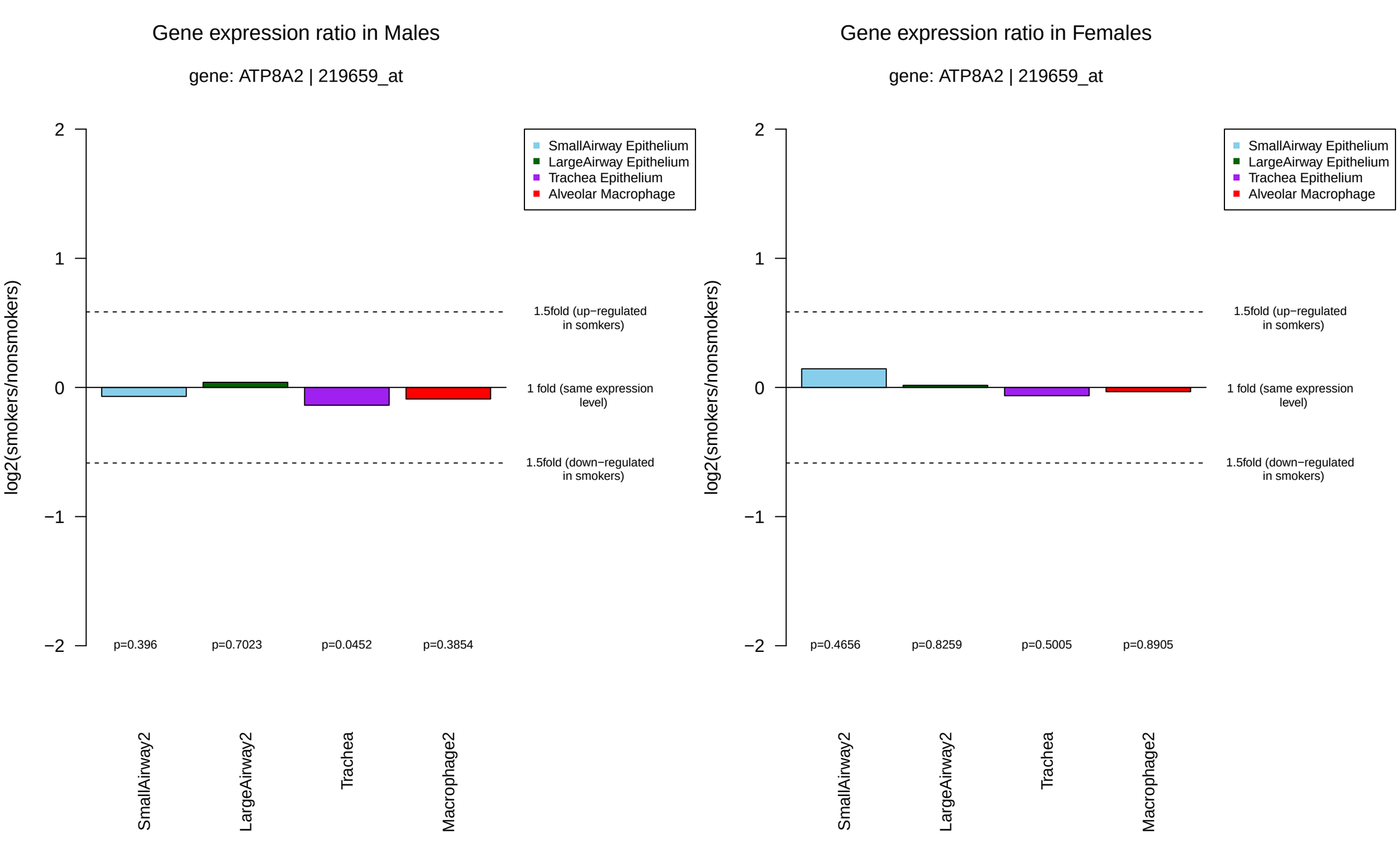
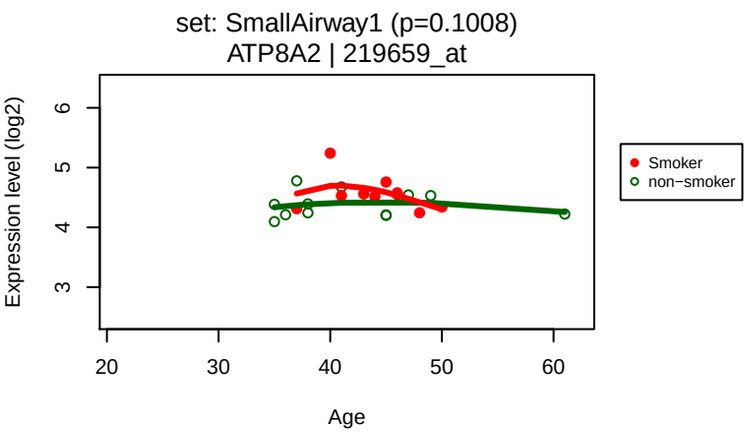
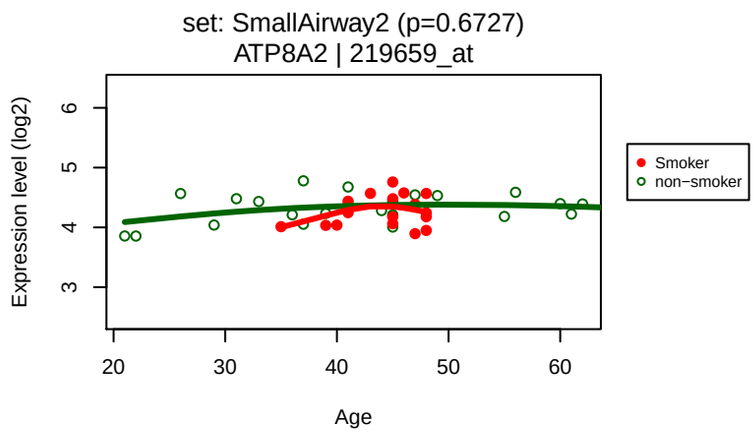
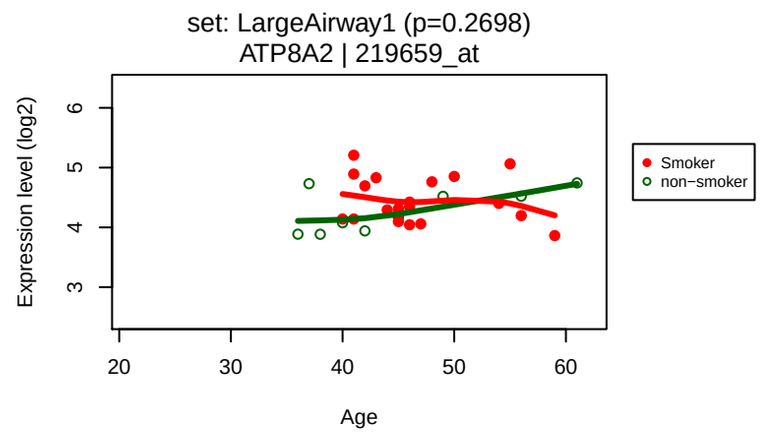

Gene expression of smokers and nonsmokers
gene: ATP8A2 | 219659_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1008
p=0.6727
p=0.2391
p=0.2698
p=0.5339
p=0.0356
p=0.3182
p=0.4313
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

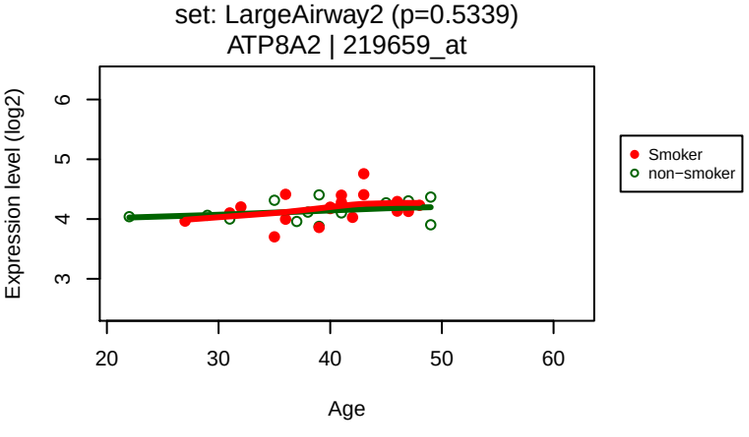

Gene expression ratio (smokers/nonsmokers)
gene: ATP8A2 | 219659_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1008
p=0.6727
p=0.2391
p=0.2698
p=0.5339
p=0.0356
p=0.3182
p=0.4313
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

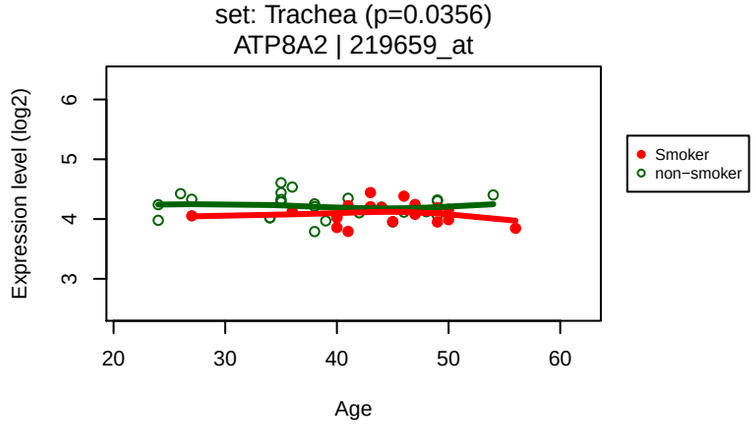

Gene expression ratio in Males
gene: ATP8A2 | 219659_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.396
p=0.7023
p=0.0452
p=0.3854
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP8A2 | 219659_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4656
p=0.8259
p=0.5005
p=0.8905
SmallAirway2
LargeAirway2
Trachea
Macrophage2

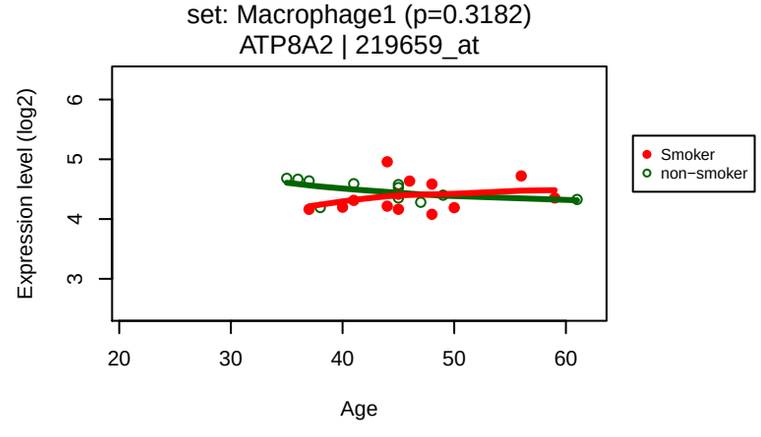

set: SmallAirway1 (p=0.1008)
ATP8A2 | 219659_at
Expression level (log2)
Age
Smoker
non−smoker

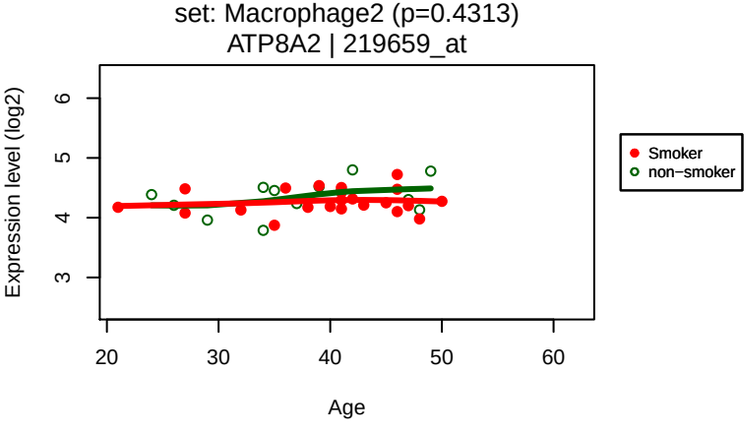

set: SmallAirway2 (p=0.6727)
ATP8A2 | 219659_at
Expression level (log2)
Age
Smoker
non−smoker

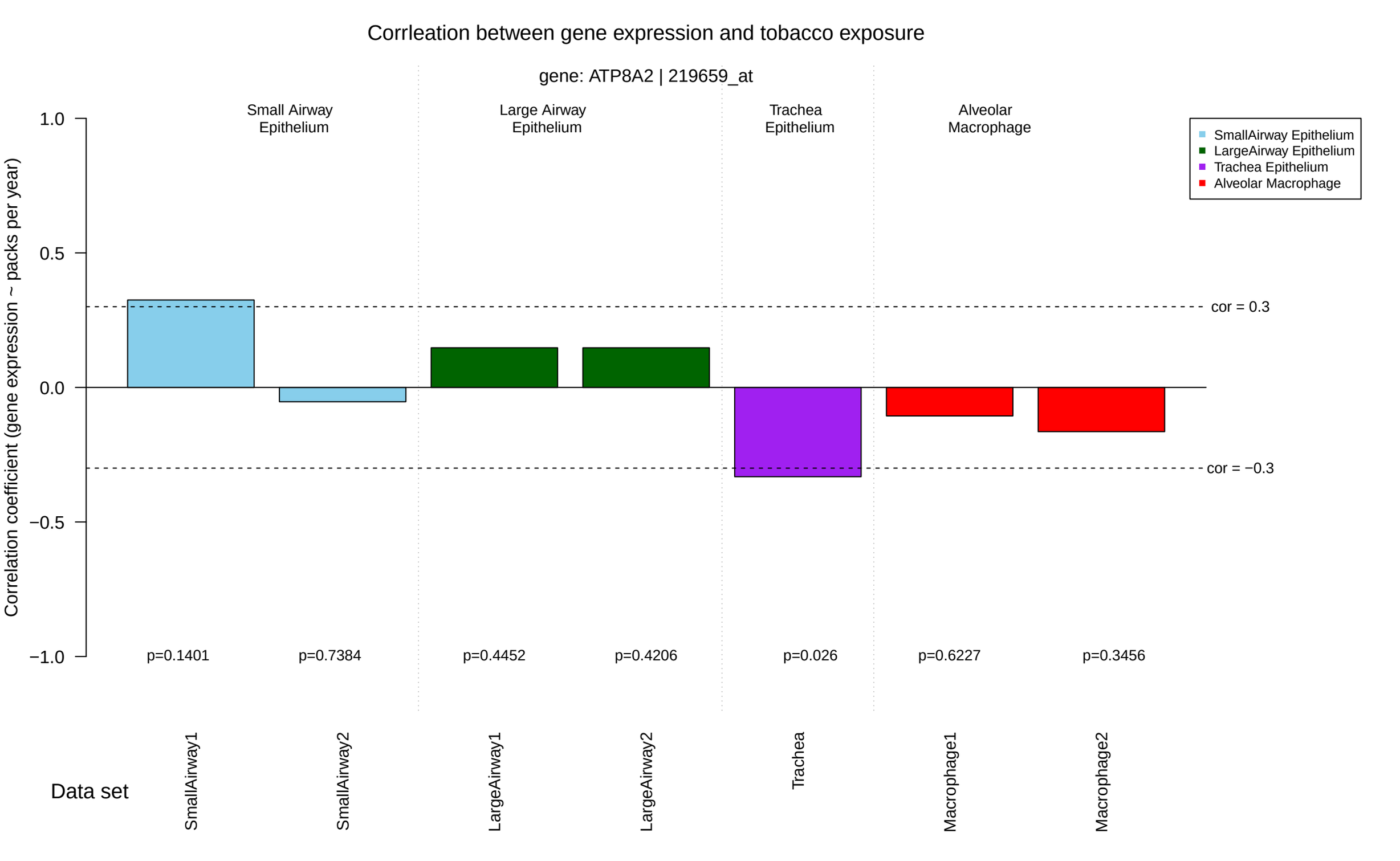

set: LargeAirway1 (p=0.2698)
ATP8A2 | 219659_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5339)
ATP8A2 | 219659_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0356)
ATP8A2 | 219659_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3182)
ATP8A2 | 219659_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4313)
ATP8A2 | 219659_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP8A2 | 219659_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1401
p=0.7384
p=0.4452
p=0.4206
p=0.026
p=0.6227
p=0.3456
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2